22PH202 – Electromagnetism and Modern Physics\

Unit II – Magnetism

2.4Gauss law

Gauss law states that the total electric flux out of a closed surface is equal to the charge enclosed divided by the permittivity. The electric flux in an area is defined as the electric field multiplied by the area of the surface projected in a plane and perpendicular to the field.

According to Gauss law, the total flux linked with a closed surface is $1/\varepsilon_0$ times the charge enclosed by the closed surface.

$$\oint \vec{E}.\vec{ds} = \frac{1}{\epsilon_0} q$$

In simple words, the Gauss theorem relates the 'flow' of electric field lines (flux) to the charges within the enclosed surface. If no charges are enclosed by a surface, then the net electric flux remains zero.

This means that the number of electric field lines entering the surface equals the field lines leaving the surface.

The electric flux from any closed surface is only due to the sources (positive charges) and sinks (negative charges) of the electric fields enclosed by the surface. Any charges outside the surface do not contribute to the electric flux. Also, only electric charges can act as sources or sinks of electric fields. Changing magnetic fields, for example, cannot act as sources or sinks of electric fields.

2.4.1 Electric Field Due to Infinite Wire – Gauss Law Application

Consider an infinitely long line of charge with the charge per unit length being λ. We can take advantage of the cylindrical symmetry of this situation. By symmetry, the electric fields all point radially away from the line of charge, and there is no component parallel to the line of charge.

We can use a cylinder (with an arbitrary radius (r) and length (l)) centred on the line of charge as our Gaussian surface.

As you can see in the above diagram, the electric field is perpendicular to the curved surface of the cylinder. Thus, the angle between the electric field and area vector is zero and cos $\theta = 1$.

The top and bottom surfaces of the cylinder lie parallel to the electric field. Thus, the angle between the area vector and the electric field is 90 degrees, and cos θ = 0.

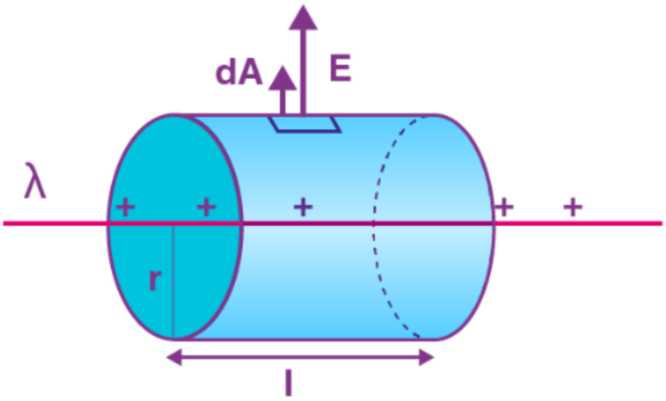

Figure 11.Electric field due to infinite wire – Gauss law application

According to Gauss's Law

$$\Phi = \frac{q}{\varepsilon_0}$$

$$E \times 2\pi r l = \frac{\lambda l}{\varepsilon_0}$$

$$E = \frac{\lambda}{2\pi\varepsilon_0 r}$$

2.4.2 Electric Field due to Infinite Plate Sheet

Let us consider an infinite plane sheet, with surface charge density σ and cross-sectional area A. The position of the infinite plane sheet is as below:

The direction of the electric field due to an infinite charge sheet is perpendicular to the plane of the sheet. Let us consider a cylindrical Gaussian surface, whose axis is normal to the plane of the sheet. We can evaluate the electric field E from Gauss's Law as according to the law:

$$\Phi = \frac{q}{\varepsilon_0}$$

Froma continuous charge distribution charge q will be the charge density (σ) times the area (A). Talking about net electric flux, we will consider electric flux only from the two ends of the assumed Gaussian surface. We can attribute it to the fact that the curved surface area and an electric field are normal to each other, thereby producing zero electric flux. So the net electric flux is

$$\Phi = EA - (-EA)$$

$$\Phi = 2EA$$

$$2EA = \frac{\sigma A}{\varepsilon_0}$$

The term A cancels out which means electric field due to an infinite plane sheet is independent of cross-sectional area A and equals to:

$$E = \frac{\sigma}{2\varepsilon_0}$$

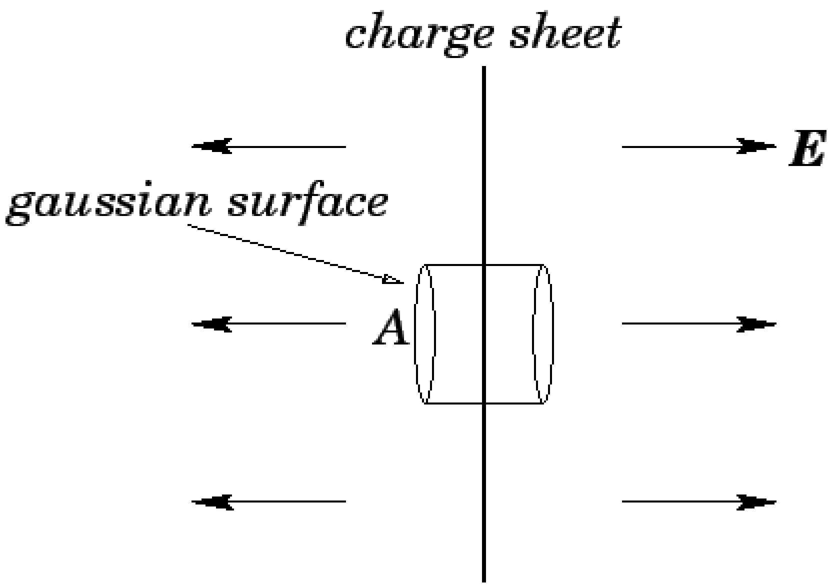

Figure 12. Electric Field due to Infinite Plate Sheet

2.4.3 Electric Field due to Thin Spherical Shell

Let us consider a thin spherical shell of surface charge density σ and radius "R". By observation, we can see that the shell has spherical symmetry. Therefore, we can evaluate the electric field due to the spherical shell in two different positions:

- ✓ Electric field outside the spherical shell
- ✓ Electric field inside the spherical shell

Let us look at these two cases in greater detail.

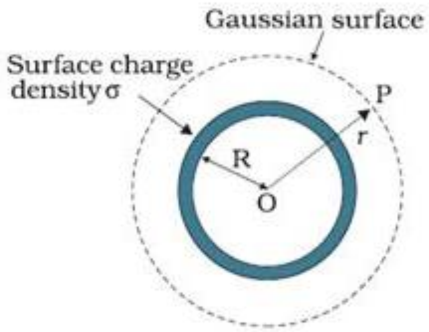

Figure 13. Electric Field due to thin spherical shell

2.4.3.1 Electric Field Outside the Spherical Shell

To find electric field outside the spherical shell, we take a point P outside the shell at a distance r from the centre of the spherical shell. By symmetry, we take Gaussian spherical surface with radius r and centre O. The Gaussian surface will pass through P, and experience a constant electric field E all around as all points are equally distanced “r’’ from the centre of the sphere. Then, according to Gauss’s Law:

$$\Phi = \frac{q}{\varepsilon_0}$$

$$\vec{E} = \frac{kq}{r^2}\,\hat{r}$$

2.4.3.2 Electric Field Inside the Spherical Shell

To evaluate electric field inside the spherical shell, let’s take a point P inside the spherical shell. By symmetry, we again take a spherical Gaussian surface passing through P, centered at O and with radius r. Now according to Gauss’s Law

$$\Phi = \frac{q}{\varepsilon_0}$$

The net electric flux will be $E \times 4\pi r^2$.